

- **strictly dominated strategy:** $u_i(s_i, s_{-i}) < u_i(s'_i, s_{-i})$ for all $s_{-i} \in S_{-i}$.
- **best-response correspondence:** $R_i(s_{-i}) := \operatorname{argmax}_{s_i \in S_i} u_i(s_i, s_{-i})$.
- **Nash equilibrium:** $s_i^* \in R_i(s_{-i}^*) \iff u_i(s_i^*, s_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, s_{-i}^*)$ for all i .
- Define $G(R_i) := \{(s_i, s_{-i}) : s_i \in R_i(s_{-i}), s_{-i} \in S_{-i}\}$ as the **graph** of R_i , then $\bigcap_{i=1}^n G(R_i)$ are exactly all the Nash equilibria.
- 2-player games: (s_1, s_2) is a Nash equilibrium iff $s_1 \in R_1(s_2)$ and $s_2 \in R_2(s_1)$.
- If $(s_1^*, s_2^*, \dots, s_n^*)$ are a Nash equilibrium, then no s_i^* can be eliminated by IESDS. **Proof: suppose s_j is the first to be eliminated.**
- If IESDS results in a single set of strategies, they are the unique Nash equilibrium of the game. **Proof: uniqueness is trivial. For existence, suppose $s_i^* \notin R_i(s_{-i}^*) \neq \emptyset$ for some i .**
- **Mixed strategy:** $p_i := (p_{ij})_{j=1}^k$ s.t. $P(\text{Player } i \text{ plays } s_{ij}) = p_{ij}$.
- **Expected payoff for pure strategy:**

$$v_i(x, p_j) := \sum_{n=1}^N p_{jn} u_i(x, s_{jn}).$$

- **Expected payoff for mixed strategy:**

$$v_i(p_i, p_j) := \sum_{m=1}^M p_{im} \sum_{n=1}^N p_{jn} u_i(s_{im}, s_{jn}).$$

- If s_{ik} is eliminated by IESDS, then $p_{ik}^* = 0$.
- **Nash's Theorem:** at least one Nash equilibrium exists for any finite game if mixed strategies are allowed.
- **Market clearing price:** the price at which the demand is equal to the supply.
- **Demand-side reservation price:** the lowest price at which the demand is 0.
- **Cournot Model of Duopoly:**

- Demand: $Q(P) = ka - kP$ where a is the reservation price.
- Price: $P(Q) = \begin{cases} 0 & \text{if } Q \geq ka \\ a - \frac{1}{k}Q & \text{otherwise} \end{cases}$.
- Profit: $\pi_i(q_i, q_j) = \begin{cases} (a - \frac{1}{k}q_i - \frac{1}{k}q_j - c) q_i & \text{if } Q < ka \\ -cq_i & \text{if } Q \geq ka \end{cases}$.
- Best response: $R_i(q_j) = \begin{cases} \frac{1}{2}(ka - q_j - kc) & \text{if } q_j \leq k(a - c) \\ 0 & \text{otherwise} \end{cases}$.

- **Bertrand Model of Duopoly:**

- Demand: $q_i(p_i, p_j) = ka - kp_i + bp_j$ where a is the reservation price and b is the substitutability coefficient.
- Profit: $\pi_i(p_i, p_j) = (ka - kp_i + bp_j)(p_i - c)$.
- Best response: $R_i(p_j) = \frac{1}{2k}(ka + bp_j + kc)$.
- Nash equilibrium: $p_1^* = p_2^* = \frac{ka+kc}{2k-b}$, i.e., the model makes sense iff $b > 2k$.
- Homogeneous product Nash equilibrium: $p_1^* = p_2^* = c$.

- **Problem of the commons:**

- Unit utility given total consumption: $v(x)$ is **strictly concave** for $x < X_{\max}$ and $v(x) = 0$ for $x \geq X_{\max}$.
- Payoff: $u_i(x_i, x_{-i}) := x_i v(X) - cx_i$.
- Nash equilibrium solves $v(X^*) + x_i^* \frac{dv}{dx_i} \Big|_{x_i=x_i^*} - c = 0$.
- If $v(x) = a - x^2$ for $x < X_{\max}$, then $x_i^* = \sqrt{\frac{a-c}{n^2+2n}}$.
- Social welfare: $w(X) := Xv(X) - Xc$.
- If x^* optimises w , then $x^* < X^*$. **Proof: if $x^* \geq X^*$, then $v(X^*) \geq v(x^*)$ and $0 > v'(X^*) \geq v'(x^*)$. Use this to show $v(X^*) + \frac{X^*}{n} v'(X^*) > v(x^*) + x^* v'(x^*)$. Use stationary points to show $w'(x^*) = 0$ and $u'_i(x_i^*) = 0$.**

- **Final-offer arbitration:**

- Union offers w_u and firm offers w_f .
- Arbitrator expectation $x \sim f$ with $P(x < x_m) = F(x_m) = \frac{1}{2}$.
- Expected wage: $\phi(w_f, w_u) = \mathbb{E}[w] = w_f F\left(\frac{w_f + w_u}{2}\right) + w_u \left(1 - F\left(\frac{w_f + w_u}{2}\right)\right)$.
- Best responses:

$$R_f(w_u) = \operatorname{argmin}_{0 \leq w_f < w_u^*} \phi(w_f, w_u^*), \quad R_u(w_f) = \operatorname{argmax}_{w_u > w_f^*} \phi(w_f^*, w_u).$$

- Nash equilibrium: $w_f^* = x_m - \frac{1}{2f(x_m)}$, $w_u^* = x_m + \frac{1}{2f(x_m)}$.

- **Backward induction:**

- Define $b_j(b_1) := R_j(b_1, \dots, b_{j-1})$ as player j 's best action given player 1's action.
- $a_i^* := \operatorname{argmax}_{a_1 \in A_1} u_1(a_1, b_2(a_1), \dots, b_n(a_1))$ and $a_i^* := b_i(a_1^*)$.

- **Extensive-form game** specifies:

1. the players in the game;
2. when each player takes a move;
3. each player's strategy space;
4. each player's knowledge of the game;

5. the payoffs for each possible combination of strategies.

- **Information set:** a set of **decision nodes** joined with **dotted line in a game tree** for the same player at a turn s.t. the player cannot deduce the predecessor node.
Perfect information $\iff$ all information sets are singletons.

- **Subgame:** a game based on a subtree of a game tree s.t. its decision nodes can be partitioned by information sets.

- **Strategy in dynamic games:** a map $s_i: \prod_{j=1}^{i-1} A_j \rightarrow A_i$.

- Action of player i following strategies $\mathbf{s} := (s_j)_{j=1}^n$ is $a_i(\mathbf{s}) = s_i(a_1(\mathbf{s}), \dots, a_{i-1}(\mathbf{s}))$.

- **Payoff in dynamic games:** $\tilde{\mathbf{u}}(\mathbf{s}) := \mathbf{u}(a_1(\mathbf{s}), a_2(\mathbf{s}), \dots, a_n(\mathbf{s}))$.

- $(a_1^*, R_2(a_1^*))$ is a backward induction outcome $\implies (a_1^*, R_2)$ is a Nash equilibrium.
Converse is not true in general.

- **Subgame perfect Nash equilibrium** is a Nash equilibrium in every subgame.

- **Stackelberg Model of Duopoly:**

– Dominate firm 1 and follower firm 2.

– Price: $P(Q) = \begin{cases} a - \frac{1}{k}Q & \text{if } Q < ka \\ 0 & \text{if } Q \geq ka \end{cases}$.

– $R_2(q_1) = \begin{cases} \frac{1}{2}(ka - q_1 - kc) & \text{if } q_1 < k(a - c) \\ 0 & \text{otherwise} \end{cases}$.

	Cournot	Stackelberg
(q_1^*, q_2^*)	$(\frac{ka-kc}{3}, \frac{ka-kc}{3})$	$(\frac{ka-kc}{2}, \frac{ka-kc}{4})$
P	$c + \frac{a-c}{3}$	$c + \frac{a-c}{4}$
(π_1^*, π_2^*)	$(\frac{k^2(a-c)^2}{9}, \frac{k^2(a-c)^2}{9})$	$(\frac{k^2(a-c)^2}{8}, \frac{k^2(a-c)^2}{16})$
$\pi_1^* + \pi_2^*$	$\frac{2k^2(a-c)^2}{9}$	$\frac{3k^2(a-c)^2}{16}$

- **Three-period bargaining:** if the players will receive (u_1, u_2) in period $i + 1$, then player i should offer δu_j and player j should accept.

- In a three-period bargaining game, player 1 will offer $\begin{cases} s_1(1) &= 1 - \delta(1 - \delta\bar{s}_1) \\ s_2(1) &= \delta(1 - \delta\bar{s}_1) \end{cases}$ and player 2 will accept the offer.

- **Infinite-horizon bargaining:** if $(\bar{s}_1, \bar{s}_2)$ are final offers received in backward induction, then by collapsing stage 3 onwards, the game is equivalent to a 3-period bargaining with $\begin{cases} \bar{s}_1 &= 1 - \delta(1 - \delta\bar{s}_1) \\ \bar{s}_2 &= \delta(1 - \delta\bar{s}_1) \end{cases}$ which yields $\bar{s}_1 = \frac{1}{1+\delta}$.

- **Multi-stage game:** dynamic games which can be partitioned into static stage games. The game is **repeated** if every stage is the same.

- If the stage game of a repeated game has a unique Nash equilibrium, then the repeated game has a unique subgame perfect Nash equilibrium.

- **Present value:** $\sum_{i=1}^{\infty} \delta^{i-1} \pi_i$ where π_i is the stage i payoff and $\delta \in (0, 1)$ is the **discount factor**.

- If the stage game with left (not cooperate) and right (cooperate) strategies is

$$\begin{bmatrix} & \text{L} & \text{R} \\ \text{L} & (1, 1) & (5, 0) \\ \text{R} & (0, 5) & (4, 4) \end{bmatrix},$$

then (NC, NC) is always a Nash equilibrium and (T, T) is a Nash equilibrium iff $\delta \geq \frac{1}{4}$.

Proof: show that T is the best response for T , i.e., if player 1 plays T , then player 2 will end up playing T .